MoodTunes AI

Emotion-Aware Bollywood Music Recommendation System

Presented By: Priyanshu Vairale
Course: BCA (AI, ML & Data Science)
Technology: Artificial Intelligence | Computer Vision | NLP

Tagline:
"*Transforming Emotions into Music Experiences*"

Introduction

Music has a significant impact on human emotions and mental well-being. People often listen to music according to their mood, but selecting the right song can be time-consuming.

MoodTunes AI is an intelligent music recommendation system that identifies a user's mood through:

- Facial Expression Recognition
- Text-Based Sentiment Analysis

Based on the detected emotion, the system recommends suitable Bollywood songs instantly.

Problem Statement

Current music platforms primarily rely on:

- Listening history
- Trending songs
- User playlists

However, they often fail to understand the user's current emotional state.

Challenges

- No real-time emotion detection

- Manual song searching
- Generic recommendations
- Lack of personalized emotional connection

Need

An AI system that understands emotions and recommends music accordingly.

Objectives of MoodTunes AI

Primary Objectives

☑ Detect emotions from facial expressions

☑ Analyze sentiment from textual input

☑ Recommend Bollywood songs based on detected mood

☑ Provide an easy-to-use graphical interface

☑ Improve user experience through personalized recommendations

Expected Outcome

Deliver mood-specific song suggestions in real-time.

System Architecture

Face-Based Recommendation

User Face Input ↓ OpenCV Camera ↓ DeepFace Emotion Analysis ↓ Mood Detection ↓ Song Recommendation ↓ Display Result

Text-Based Recommendation

Text Input ↓ TextBlob Sentiment Analysis ↓ Mood Detection ↓ Song Recommendation ↓ Display Result

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Technologies Used

Component	Technology
Programming Language	Python
Emotion Detection	DeepFace
Computer Vision	OpenCV
Sentiment Analysis	TextBlob
Data Handling	Pandas
Data Storage	CSV
User Interface	Tkinter
Machine Learning	Deep Learning Models

Advantages

- Open Source
- Fast Processing
- Easy Integration
- Real-Time Execution

How MoodTunes AI Works

Method 1: Face Detection

1. User clicks "Recommend from Face"
2. Webcam captures image
3. DeepFace analyzes facial expressions
4. Dominant emotion is identified
5. Matching Bollywood song is recommended

Method 2: Text Input

1. User enters text
2. TextBlob calculates sentiment polarity
3. Mood is identified
4. Recommendation engine selects appropriate song

Mood-Based Song Dataset

Mood	Song
Happy	Badtameez Dil
Sad	Bekhayali
Angry	Saadda Haq
Romantic	Tum Hi Ho
Spiritual	Hanuman Chalisa
Energetic	Chaiyya Chaiyya
Emotional	Kal Ho Naa Ho
Motivational	Zinda

Recommendation Logic

Mood Detection → Mood Mapping → Song Recommendation

Future Improvements

Short-Term Enhancements

- Increase Bollywood song database
- Better mood accuracy
- Enhanced GUI design

Advanced Features

🎵 Spotify Integration

🎵 YouTube Music Integration

🎵 Voice Emotion Recognition

🎵 AI Playlist Generation

🎵 Multilingual Song Recommendations

🎵 Mobile Application

🎵 Personalized Recommendation Engine

Research Potential

Can be expanded into an AI-powered emotional wellness assistant.

Conclusion

MoodTunes AI successfully combines:

- Artificial Intelligence
- Computer Vision
- Natural Language Processing
- Music Recommendation Systems

The project provides personalized Bollywood song recommendations based on a user's emotional state, creating a smarter and more engaging music experience.

Key Achievement

"Understanding Human Emotions Through AI and Converting Them Into Personalized Music Recommendations."

Thank You

Questions & Discussion

MoodTunes AI "*Feel Your Mood, Hear Your Music*" 🎵